

Rithish Murugan

AI Full Stack Software Engineer

+1 (773) 930-2475 | rithishmurugan52@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

AI Full Stack Software Engineer with **4+ years of experience** building **healthcare AI platforms, backend services**, and **distributed enterprise systems**. Progressed from **Java, Spring Boot**, and supply-chain microservices at Hexaware Technologies, through **Python, ETL, SQL**, and **REST API** development at Virtualan Software LLC, to clinician-facing **GenAI, RAG, React, FastAPI**, and cloud-native healthcare integrations at Abridge. Brings strong ownership, structured problem solving, and cross-functional collaboration while delivering secure, reliable production systems.

SKILLS

Programming Languages: Python, Java, SQL, TypeScript

Gen AI & Agents: LLMs, GPT-4o, Claude, Llama 3, LangChain, LangGraph, RAG, Agentic AI, Function Calling, Pinecone

Backend Engineering & APIs: FastAPI, Spring Boot, REST APIs, Microservices, Event-Driven Architecture, Asynchronous Processing, Hibernate, JPA, Swagger/OpenAPI, Performance Optimization

Frontend Development: React, AI Copilot Interfaces, Citation Review Interfaces, Feedback Capture Workflows

AI Evaluation & Safety: Retrieval Evaluation, Groundedness, Citation Validation, HITL, PII Safeguards, AI Governance

Data, Databases & Streaming: PostgreSQL, Snowflake, Amazon Redshift, Amazon S3, Apache Kafka, Redis, ETL, Incremental Loading, Dead-Letter Queues, Distributed Caching, Data Validation

Cloud, DevOps & Observability: AWS, EKS, Docker, Kubernetes, Terraform, Jenkins, CI/CD, CloudWatch, Git, Automated Builds, Centralized Logging, Health Checks

Testing & Software Quality: JUnit, Mockito, Integration Testing, Automated Testing, Postman, API Validation, Request Validation, Exception Handling, Code Reviews, Security Checks

Machine Learning: XGBoost, Scikit-learn, Predictive Modeling, Risk Prioritization

Healthcare & Security: EHR, FHIR, HL7, RBAC, Clinical Data Integration, Audit Logging, Healthcare Compliance

Architecture & Delivery: System Design, LLD, API Contracts, Service Boundaries, Factory Pattern, Requirements Analysis, Agile, Release Management, Production Support

EXPERIENCE

Abridge, AI Full Stack Software Engineer

Jan 2025 – Present | Remote, USA

- Architect scalable healthcare **microservices** integrating five clinical domains through **Python, FastAPI, FHIR, HL7**, and **PostgreSQL**, centralizing 400K+ patient records while documenting APIs, security requirements, and compliance standards.
- Orchestrate event-driven clinical data services with **Apache Kafka, Redis, Amazon S3**, and **Snowflake**, implementing retries, dead-letter handling, caching, and validation to improve reliability and data consistency.
- Deliver clinician-facing **AI copilot APIs** and review interfaces using **FastAPI, React, TypeScript, LangChain, RAG, GPT-4o, Claude**, and **Llama 3**, enabling citation review, feedback capture, and secure approval.
- Operationalize **LangGraph workflows** with **function calling, asynchronous orchestration**, and **human review**, validating retrieval quality, groundedness, citations, and **PII safeguards** across 50K+ monthly clinical interactions.
- Coordinate with physicians, product leaders, and compliance teams to validate **XGBoost** and **Scikit-learn** risk-prioritization models, defining performance thresholds, clinician-review criteria, deployment controls, and production decision boundaries.
- Optimize cloud-native delivery with **AWS EKS, Docker, Kubernetes, Terraform, CI/CD**, and **CloudWatch**, establishing health checks, audit logging, and controls while sustaining 99.99% uptime and reducing costs 25%.

Virtualan Software LLC, Software Engineer

May 2024 – Aug 2024 | Chicago, IL

- Developed scalable backend **ETL pipelines** using **Python, SQL, AWS S3**, and **Amazon Redshift**, transforming 10K+ daily records with **data validation** and **incremental loading** to reduce reporting delays for analytics stakeholders reliably.
- Built and documented **REST APIs** using **Python, SQL, Postman**, and **Swagger/OpenAPI**, adding **request validation** and **automated testing** that increased coverage by 35% and streamlined cross-functional backend integrations for product teams.
- Implemented backend controls **RBAC, data validation**, and **peer code reviews** while coordinating **Agile delivery** and release workflows, reducing preventable defects and supporting stable releases throughout the four-month engagement.

Hexaware Technologies, Java Developer

Mar 2021 – Jul 2023 | Chennai, India

- Analyzed inventory, order, and shipment requirements with product owners and operations stakeholders, translating **user stories** into **API contracts** and delivery plans that reduced development rework.
- Designed **low-level architecture** for **Java microservices** using **Spring Boot, Hibernate, PostgreSQL**, and **Factory patterns**, defining service boundaries and database schemas supporting eight regional warehouses.
- Developed secure **REST APIs** and event-driven business logic with **Java, Spring Boot, Kafka, Redis, JPA**, and **PostgreSQL** for inventory updates, improving average service response time by 18%.
- Strengthened code quality through **JUnit, Mockito, integration testing, API validation**, and structured **exception handling**, increasing critical-service test coverage from 62% to 78% before releases.
- Reviewed peer code through **Git**, enforced **Java coding standards**, and integrated **Jenkins CI/CD pipelines** with automated builds, tests, and security checks, improving release consistency across shared services.
- Containerized **Spring Boot** services with **Docker**, deployed them through **Kubernetes** on **AWS**, and monitored logs with **CloudWatch**, resolving production issues while maintaining reliable warehouse operations.

EDUCATION

Master of Science in Computer Science | Illinois Institute of Technology

Aug 2023 – May 2025 | Chicago, IL